Bayesian Reasoning and Decisions

Unit 11: Updating Beliefs, and Acting on Them

Stat 220 | Statistics for Data Science

Where this fits

- Everything so far asked: if the null were true, how surprising is this data?
- The question people actually ask is the reverse: given this data, what should I now believe, and what should I do?
- That reversal is Bayes' rule, which you built in Unit 6 for medical tests. Here we point it at the unknown quantity itself, and then at the decision.

Principle: the question you were always trying to answer

A p-value is $P(\text{data} \mid \text{hypothesis})$. A posterior is $P(\text{hypothesis} \mid \text{data})$. People misread the first as the second constantly, because the second is what they wanted.

What the game teaches

- Belief is a number that moves, not a verdict you reach.
- How far it moves depends on two things only: where you started, and how much more likely the data is under one hypothesis than the other.
- That ratio, $0.7/0.3$ here, is the **evidence**. Everything else is bookkeeping.

In practice: the shape of this module

"How confident are you that the new model is better?" A Bayesian answer gives a number, and can say what it would take to change it.

What this unit gives you

Things to know

- probability as a degree of belief, and when that reading is the useful one
- prior, likelihood, and posterior, and how they combine
- why the prior matters with little data and washes out with plenty
- what a credible interval says, and how it differs from a confidence interval
- why Bayesian estimates shrink small groups automatically
- how to turn a posterior into a decision using costs

Mistakes to stop making

- pretending you have no prior when you obviously do
- letting a strong prior drive a conclusion the data cannot support
- reading a 95% confidence interval as a 95% probability statement
- treating a high posterior probability as proof rather than as a bet
- forgetting that Bayes assumes your *model* is right
- reporting a posterior without saying which prior produced it

Prior, likelihood, posterior

$$\underbrace{P(\theta \mid \text{data})}_{\text{posterior}} \propto \underbrace{P(\text{data} \mid \theta)}_{\text{likelihood}} \times \underbrace{P(\theta)}_{\text{prior}}$$

- The **prior** is what you believed before: past experiments, industry benchmarks, physical limits.
- The **likelihood** is the same object from the estimation module: how well each parameter value explains the data.
- The **posterior** is a full distribution over the unknown, not a point. Every question you have is answered by an area under it.

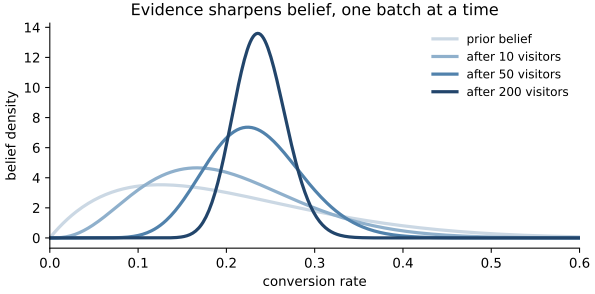
Updating a rate, with the actual numbers

Conjugate shortcut: with a Beta(a, b) prior, s successes and f failures give a Beta$(a + s,\ b + f)$ posterior. The prior acts like a imaginary successes and b imaginary failures you saw before the study.

after	data so far	posterior	mean	95% interval
nothing	none	Beta(2, 8)	0.200	0.03 to 0.48
10 visitors	2 of 10	Beta(4, 16)	0.200	0.07 to 0.40
50 visitors	12 of 50	Beta(14, 46)	0.233	0.14 to 0.35
200 visitors	48 of 200	Beta(50, 160)	0.238	0.18 to 0.30

- The prior contributes 10 imaginary observations. After 200 real ones it is outvoted 20 to 1, and the interval has shrunk by two thirds.

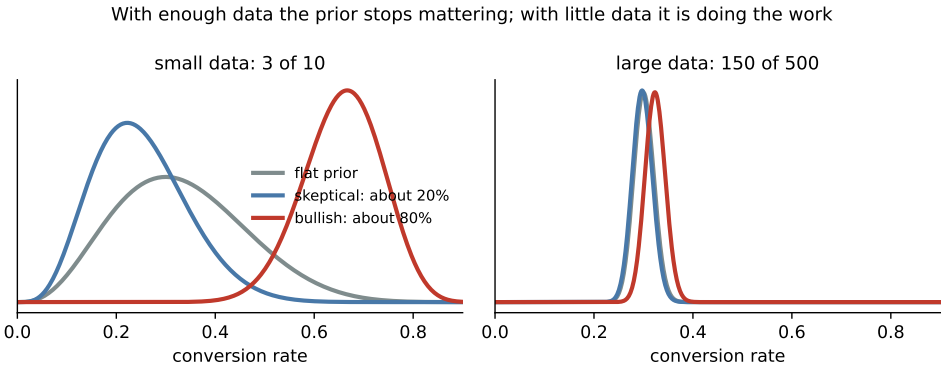
Belief sharpening, batch by batch



Estimating a conversion rate, starting from a prior centered near 20%.

- Each batch of visitors narrows the distribution.
- Nothing special happens at any threshold. There is no moment of "significance," just steadily less uncertainty.

Does the prior take over the answer?



Left: with 10 observations, three reasonable priors give three different answers. Right: with 500, they are indistinguishable. The prior is a real assumption exactly when the data is thin, which is exactly when you needed one.

"Isn't the prior just making things up?"

- You already have a prior. When a colleague reports a 400% conversion lift, you do not believe it, and that disbelief is a prior doing its job.
- The Bayesian move is to write it down where others can argue with it, instead of applying it silently after seeing the result.
- The professional practice: use a weak prior, then run a **sensitivity analysis**, reporting how the conclusion changes under a skeptical and an optimistic prior.

You may be asked: how to answer the subjectivity objection

"Priors are assumptions, and so are the model, the independence, and the choice of test. I would rather state mine explicitly and show the answer under several of them."

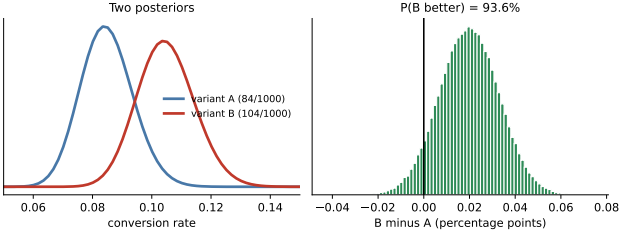
Credible intervals versus confidence intervals

- **Credible interval:** given this data, prior, and model, there is a 95% probability the parameter is in here. That is the sentence everyone wants.
- **Confidence interval:** the procedure that produced this interval captures the truth 95% of the time in repeated sampling. The probability is a property of the method.
- In practice with plenty of data and a weak prior, the two are usually very close. The difference matters most when data is scarce or prior knowledge is strong.

Trap: the universal misreading

Nearly everyone reads a confidence interval as a credible interval. If that is the statement you want, say so, use a prior, and earn it.

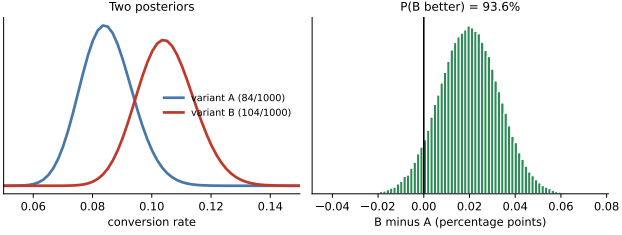
A Bayesian A/B test



1,000 visitors each: A converts 84, B converts 104.

- $P(\text{B better than A}) = \mathbf{93.6\%}$.
- The same data gives a frequentist $p = 0.125$, which most people read as "not significant."

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Both are correct. They answer different questions, and only one of them is the question the product manager asked.

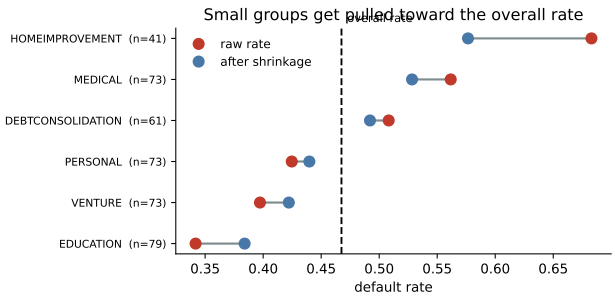
From posterior to decision: expected loss

- Do not stop at "93.6% likely better." Ask what it costs to be wrong.
- **Expected loss** of shipping B: average, over the posterior, of how much worse B is whenever it is worse. Here that is about **0.04 percentage points** of conversion.
- If 0.04 points is negligible relative to the upside, ship. The decision rule is a cost comparison, not a threshold on probability.

Principle: the deliverable is an action with its expected cost

"Ship B. The chance we are wrong is 6%, and the expected cost if we are is tiny compared with the expected gain" is a complete answer.

Shrinkage, for free



Default rates by loan purpose in a 400-loan sample.

- Small categories have wild raw rates.
- A prior centered on the overall rate pulls them in, by an amount that depends on their sample size.

This is the leaderboard fix from the estimation module, arising automatically.

Your turn #1

Your turn: how much do you believe it?

A colleague reports an A/B test on a checkout button: **240% lift** in conversion, $p = 0.03$, based on 60 visitors per arm.

With a neighbor: what is your posterior belief that the true lift exceeds 100%? What prior are you using, and what would change your mind?

Your turn #1: answer

- Your prior, built from every A/B test ever run, says real button changes move conversion by a few percent, not by 240%.
- With 60 visitors per arm the likelihood is very flat, so it barely moves that prior. The posterior stays near "a small effect, if any."
- This is the winner's curse in Bayesian language: extreme estimates from tiny samples are mostly noise, and a sensible prior discounts them automatically.
- What would change your mind: a preregistered replication at ten times the sample size.

Principle: extraordinary claims need extraordinary likelihood ratios

The more surprising the claim, the more evidence it takes to move you. That is arithmetic, not stubbornness.

Your turn #3

Your turn: how much would this move you?

Your company's landing pages historically convert between 6% and 14%. A new page is tested and converts **4 of 10** visitors, which is 40%.

With a neighbor: pick a prior that encodes the company history, work out roughly where the posterior lands, and decide what you would tell the person who wants to ship it today.

Your turn #3: answer

- A prior centered near 10% with real weight, say Beta(10, 90), encodes "about 10%, worth roughly 100 past observations."
- Posterior: Beta$(10+4,\ 90+6)$ = Beta(14, 96), a mean near **12.7%**. The spectacular 40% barely moves the needle, because 10 visitors is almost no evidence.
- What to say: "This is encouraging but it is 10 people. Our best estimate is still about 13%. Run it to a few thousand visitors before we decide."

Principle: the winner's curse, in Bayesian clothing

A sensible prior automatically discounts extreme results from tiny samples. That is not conservatism. It is what the arithmetic says once you admit you knew something beforehand.

When to stop a test

- In the frequentist frame, peeking at a p-value and stopping when it dips below 0.05 inflates the false-positive rate badly. You saw this in Part 1.
- A posterior can be looked at any time without that particular problem, because it is a statement about current belief rather than a long-run error rate.
- But it is not a free lunch: if you stop the moment the posterior crosses a threshold, you still systematically overstate the effect. Use a decision rule based on **expected loss** and a planned horizon.

Trap: "Bayesian methods let you peek"

They let you look. They do not protect you from stopping early on a lucky run and shipping an inflated estimate.

Your turn #2

Your turn: price the information

You are deciding whether to launch in a new city. Launching costs $400k. Your posterior says a 60% chance of a $1M return and a 40% chance of $0.

With a neighbor: (i) should you launch? (ii) A market study costing $50k would tell you which case you are in with high accuracy. Is it worth buying?

Your turn #2: answer

- (i) Expected return is $0.6(1{,}000) + 0.4(0) = \$600$k against a \$400k cost, so the expected profit is **+\$200k**. Launch, if you can absorb the 40% chance of losing \$400k.
- (ii) With a perfect study you launch only in the good case: expected profit $0.6 \times \$600\text{k} = \$360\text{k}$. The study is worth up to $\$360 - \$200 = \mathbf{\$160k}$, so \$50k is a good buy.

Principle: information has a computable price

The value of a study is the improvement in your expected outcome from being able to act on what it tells you. If a study cannot change your decision, its value is zero, no matter how interesting it is.

LLM check: mistake or not? (1 of 2)

LLM check: we asked an LLM to interpret an interval

We told it: "Our 95% confidence interval for the treatment effect is 2 to 8. What does that mean?"

It answered: "There is a 95% probability that the true treatment effect is between 2 and 8."

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Mistake, or no mistake?

Trap: verdict: mistake

That is the definition of a *credible* interval. A confidence interval says the procedure captures the truth 95% of the time across repeated samples. The parameter is not treated as random. The two often nearly coincide, but the sentence as written is the classic misreading.

LLM check: mistake or not? (2 of 2)

LLM check: same territory, a different answer

We told it: "We have 12 sales from a brand-new region. The conversion rate looks like 40%, way above our 9% average. Should we expand there?"

It answered: "With 12 observations that 40% carries enormous uncertainty. I would shrink it toward the 9% company average using a prior built from your other regions, which will land the estimate closer to 12–15%, and I would treat expansion as a decision under uncertainty: what does it cost if the true rate is 9%?"

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Mistake, or no mistake?

Principle: verdict: no mistake

Prior from related data, shrinkage proportional to sample size, and a decision framed by costs. That is this unit in three sentences.

The arc of a Bayesian analysis

1. **Write the unknown in words** and what decision depends on it.
2. **State a prior** from past data or domain knowledge, and say where it came from.
3. **Write the likelihood**, which is just your model of how the data arises.
4. **Compute the posterior**, by formula in simple cases and by simulation otherwise.
5. **Summarize what matters**: the credible interval, and the probability of the statements someone actually cares about.
6. **Attach costs and choose an action**, reporting the expected loss.
7. **Run a sensitivity check** under a skeptical and an optimistic prior.

Scenario: the executive who wants one number

Setup. You present a posterior. The VP says: "Just tell me, does it work or not?"

- Give the decision, not the distribution: "Yes, ship it. There is a 94% chance it is better, and if we are wrong the cost is about $8k a month, against an expected gain of $140k."
- Then offer the caveat that could change the call: "This assumes next quarter looks like last quarter. If the seasonal mix shifts, I would re-check in six weeks."
- Keep the distribution in the appendix, for the people who will ask.

Principle: translate, do not simplify

Executives are not asking you to discard uncertainty. They are asking you to convert it into a recommendation with a stated risk.

From prior to decision

1. **Write the unknown in words** and say what decision depends on it.
2. **State a prior** from past data, and say where it came from.
3. **Update.** For a rate, a Beta(a, b) prior with s successes and f failures becomes Beta$(a + s,\, b + f)$.
4. **Summarize the posterior**: the mean, and a credible interval from the 2.5 and 97.5 percentiles.
5. **Answer the question actually asked**, such as P(B beats A), by drawing from both posteriors and comparing.
6. **Attach costs** and compute the expected loss of each action.
7. **Re-run under a skeptical and an optimistic prior** and report whether the decision changes.

The traps

Trap: the ones that bite most often

- Claiming you have no prior.
- Letting a confident prior overwhelm data it should have updated on.
- Reading a confidence interval as a probability about the parameter.
- Treating a 95% posterior as certainty rather than a one-in-twenty bet.
- Forgetting that Bayes assumes the model, not just the prior.
- Stopping a test the moment the posterior crosses a line.
- Reporting a posterior without naming the prior behind it.

Ten questions to be able to answer

1. Explain prior, likelihood, and posterior with an example from your own work.
2. What is the probability variant B is better, and how is that different from a p-value?
3. Is choosing a prior unscientific? Defend your practice.
4. Credible interval versus confidence interval: state both precisely.
5. We have 12 observations from a new market. How do you estimate the rate?
6. How do you turn a posterior into a ship-or-not decision?
7. What is expected loss, and why report it alongside a probability?
8. Can you peek at a Bayesian test whenever you like? Explain carefully.
9. How much is a market study worth, and when is it worth nothing?
10. A colleague reports a 240% lift from 60 users. What is your posterior?

Mastery checklist

You have this unit when you can, from memory:

- state Bayes' rule for a parameter and name each piece
- explain when the prior matters and when it washes out
- interpret a credible interval correctly and contrast it with a confidence interval
- explain why Bayesian estimates shrink small groups
- compute the probability of a business-relevant statement from a posterior
- convert a posterior into a decision with an expected loss
- run and report a sensitivity analysis over priors

What you should be able to do with software

Nobody is asking you to memorize syntax. You are expected to know *which* procedure the question calls for, what to hand it, and what the output means. With an AI assistant open, you should be able to produce any of these and defend the result.

You should be able to	**What you would reach for**
Posterior for a rate	`stats.beta(a+s, b+f)`
Credible interval	`posterior.ppf([0.025, 0.975])`
Draw from a posterior	`posterior.rvs(200_000)`
Probability one variant beats another	`(draws_b > draws_a).mean()`
Expected loss of a decision	`np.maximum(a - b, 0).mean()`
Shrunken group estimates	`(successes + k*p_bar) / (n + k)`
Value of a study	simulate deciding with and without it, and difference the profits

If you cannot say what the output means, the fact that you produced it is worth nothing.

Where the course ends

- Part I gave you models and the discipline to distrust them. Part II gave you the machinery underneath and two applications that needed it.
- What ties them together is one move, repeated: produce a number, say how much it could have been otherwise, and say what would have to be true for it to mean what you claim.

Principle: the whole course, in one sentence

Anyone can produce a number. You have been trained to ask what would have to be true for that number to mean anything.